

AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

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Learning Engineering for Next-Generation Systems · v2.1

EDITION · 15 JUNE 2026

SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the CLOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

- I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.
- II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.
- III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

On verification. The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset’s standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE

Case 5

Operation Eagle Claw (1980). Cross-organizational capability that existed on no service’s deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.

STANDOUT SUCCESS

Case 18

U.S. Nuclear Navy / Rickover Training Model (1954–). The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

1	USS Fitzgerald & USS John S. McCain	2017	T·K·N
2	Marine Corps Training in the INDOPACOM AOR	ongoing	T·K
3	F-35 Sustainment & Maintainer Shortage	ongoing	T·K·D
4	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T·H·K
5	Operation Eagle Claw	1980	T·K
8	Patriot Missile / Dhahran	1991	D·H·K
17	GAO Weapon–System Sustainment Reviews — Operating Costs without Decision–Grade Data	2022 (GAO-22-104533)	G·K
18	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T·K·N
21	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard (series since 1968)	2020 revision	G·K
33	GIFT and the Adoption Gap	2012 – present	K·G·N
37	Navy Surface Warfare Readiness Reform	2018 – present	T·K·N

42	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks	2009 – 2014	H·K·D
54	USS Vincennes & Iran Air Flight 655	1988	H·T
64	Stanislav Petrov / 1983 False Alert	1983	H·T
138	Mark 14 Torpedo Failures	1941 – 1943	T·K·G
141	V-22 Osprey	1991 – present	T·H·N
167	9/11 Intelligence Sharing Failures	1996 – 2001	G·K

AVIATION

STANDOUT FAILURE

Case 7

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS

Case 70

Crew Resource Management & CAST (1981–). First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

6	AeroPerú Flight 603	1996	H·T
7	Boeing 737 MAX / MCAS	2018 – 2019	D·T·H
12	Boeing 737 Rudder Hardovers	1991, 1994	H·D
28	Air France Flight 447	2009	T·H
55	Kegworth / British Midland 92	1989	T·H·K
56	Asiana Airlines Flight 214	2013	T·H
57	Helios Airways Flight 522	2005	T·H
58	TransAsia Airways Flight 235	2015	T·H
63	Eastern Air Lines Flight 401	1972	H·T
68	Air Canada Chatbot Liability — Delegation Without Revocation	2022 – 2024	D·K·N
70	Crew Resource Management & CAST	1981 – present	T·H·N
72	Korean Air Safety Transformation	2000 – present	T·N
82	Colgan Air Flight 3407	2009	T
83	Atlas Air Flight 3591	2019	T·K
88	F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap	2008 – 2012	H·K·N
111	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N

118	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H-K-G
119	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson Version 7.1)	1981 – 2008 (TCAS II	H-K-G
164	NASA Saturn V Documentation	1972 – present	K
175	Singapore Airlines Safety Transformation	1980s – present	T-N

HEALTHCARE

STANDOUT
FAILURE

Case 61

EHR / CPOE Implementation (2005–). Roughly \$30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT
SUCCESS

Case 109

WHO Surgical Safety Checklist (2008–). One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

20	California Nurse Staffing Ratios — Legislating a Capability Requirement	2004 – 2010	G-K
32	ACGME 80-Hour Resident Duty-Hour Reform	2003–2017	T-K-N
34	Implementation Science in Healthcare — The 17-Year Gap	ongoing	K-G-N
59	Therac-25	1985 – 1987	H-D-G
61	EHR / CPOE Implementation	2005 – present	H-D-G
71	Keystone ICU / Pronovost Checklist	2004 – present	T-N
74	TREWS / Sepsis Watch	2018 – 2022	H-K-G
75	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff	2024 – 2025	H-K-N
79	ChatGPT in Healthcare — Hallucination Cases	2023 – present	H-D
86	Mid Staffordshire NHS Foundation Trust	2005 – 2009	G-N-K
93	Epic Sepsis Model	2017 – 2021	D-G-N
102	Medical Errors as Systemic Failure	1999 – present	T-H-N-K-G
105	Vioxx Withdrawal	1999 – 2004	G-D
109	WHO Surgical Safety Checklist	2008 – present	T-N
112	Bristol Heart Babies Reform	1984 – present	G-N
125	Annual-Screening UI Redesign + CDS at University of Missouri Health Care	2020	T-D-N
126	Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal	2019 – 2024	T-D-N

129	COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case	2022 – 2024	T·K·D
130	Radiology AI Miscalibration	2018 – present	H·K·D
132	AlphaFold — Protein Structure Prediction	2020 – present	T
136	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback	2020	T·K·D
150	Theranos	2003 – 2018	G·D
173	TeamSTEPPS	2006 – present	T·N
174	Tylenol Recall	1982	G·N

EDUCATION AT SCALE

STANDOUT FAILURE

Case 149

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS

Case 110

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

38	Cognitive Tutor / Carnegie Learning	1990s – present	T
89	Tennessee Voluntary Pre-K Study	2009–2018	G·D
90	Algorithmic Bias in Educational Predictive Analytics	ongoing	G·H·D
104	Atlanta Public Schools Cheating Scandal	2009 – 2015	G·N
110	Georgia State University Predictive Analytics	2012 – present	T·K
146	inBloom	2014	G
149	Summit Learning / Personalized Learning Rollout	2014–2019	G·T·K
171	xAPI / Total Learning Architecture — Interoperability Gap	ongoing	K·G
194	The Discipline We Build Next	ongoing	T·K·N·G·H·D

ENERGY

STANDOUT
FAILURE

Case 69

Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT
SUCCESS

Case 172

INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

11 Sago Mine Disaster 2006

N-T-K

60 Northeast Blackout 2003

H-K

69 Three Mile Island 1979

T-H

85 Deepwater Horizon 2010

T-N-K

87 Upper Big Branch Mine Explosion 2010

N-G-K

137 Texas City BP Refinery Explosion 2005

N-T-K-G

142 Davis-Besse Reactor Head Corrosion 2002

N-K-G

143 Fukushima Daiichi 2011

N-G-K

154 Camp Fire / PG&E 2018

G-N-K

172 INPO and the Nuclear Academy 1979 – present

T-K-G

GOVERNMENT

STANDOUT
FAILURE

Case 162

Australia Robodebt (2016–20). An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found “venality, incompetence and cowardice”; the scheme was linked to deaths, including suicide.

NOTE

Case 111

No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

14 Healthcare.gov Launch 2013

K-T-G

101 VA Wait-Time Scandal 2014

G-K-N

103 LIBOR Manipulation 2003 – 2012

G-N

131 Predictive Policing — PredPol 2011 – present

G-H-D

166	Bernard Madoff / SEC Failures	1992 – 2008	G-K-N
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INDUSTRIAL

STANDOUT FAILURE	Case 147	Bhopal (1984). Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.
STANDOUT SUCCESS	Case 73	Toyota Production System / Andon Cord (1950s–). Any operator can stop the line. The smallest authority–architecture intervention in the dataset, and the most durably copied.

30	Kodak Digital Camera Stagnation	1975–2012	D-K-N
73	Toyota Production System / Andon Cord	1950s – present	N-G
84	Takata Airbag Inflators	2008 – 2023	D-G
92	Volkswagen Dieselgate	2015	D-G
139	GM Ignition Switch	2002 – 2014	D-G
147	Bhopal	1984	T-K-N-G
148	Grenfell Tower	2017	G-T-K-N
153	Hyatt Regency Walkway Collapse	1981	D-G

TECHNOLOGY

STANDOUT FAILURE	Case 66	UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub–postmasters’ reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest–running operator–feedback failure in the dataset.
NOTE	Case 132	The technology dataset’s success story is filed under Healthcare: AlphaFold (Case 132), the contemporary worked example of computational capability–engineering done as a discipline.

10	Knight Capital Trading Loss	2012	D-K
31	BlackBerry Touchscreen Inertia	2007–2013	D-N-K
66	UK Post Office Horizon Scandal	1999 – 2015	G-H-K
80	AI-Augmented Coding Tools	2021 – present	T-H
91	Wells Fargo Fake Accounts	2011 – 2016	G-N

144	CrowdStrike Falcon Outage	2024	D·K·G
145	TSB Bank IT Migration	2018	H·G
151	Cambridge Analytica / Facebook	2014 – 2018	G
152	Equifax Data Breach	2017	G·K

AI IN EDUCATION

135	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models	2025	T·K·N
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K-12 EDUCATION

36	Growth-Mindset National Experiment — Heterogeneous Effects	2019	D·N·K
52	Chen et al. — Rural China AI Devices and the Equity-Direction Finding	2025	T·K·D
97	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism	2010s – 2022	G·K·N

K-12 MATHEMATICS

49	ASSISTments — National Replication and Long-Term Follow-Through	2014 – present	T·K·D
127	Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive	2007 – 2014	T·K·D

K-12 SCIENCE

51	Zhang/Scardamalia — Knowledge Building Across Three Cohorts	2009	T·K·D
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AIR TRAFFIC MANAGEMENT

25	Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change	2005	G·K·H
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ANESTHESIOLOGY

116	Anesthesia Monitoring Standards and the APSF — The Mortality Decline	1986 – present	H·K·G
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AUTONOMOUS SYSTEMS

STANDOUT FAILURE	Case 62	Uber ATG / Tempe Fatality (2018). The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.
NOTE	Case 194	No paired-intervention success in the autonomous-systems dataset. The capability engineering this category needs is still being built — see Case 194 (The Discipline We Build Next) for the open question this volume closes on.

62	Uber ATG / Tempe Fatality	2018	T·N·G·H
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78	Tesla Autopilot — Recurring Fatalities	2016 – present	T·N·G·H
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192	Cruise Robotaxi — Pedestrian Drag	2023	G·D·H
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AUTONOMOUS VEHICLES

156	Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment	2023	G·K·N
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183	Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact	2023 / 2025	G·K·N
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184	CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver	2020 – 2024	G·K·D
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AVIATION INFRASTRUCTURE

23	FAA NextGen and the ADS-B Out Transition	2003 – 2020	G·D·K
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AVIATION SAFETY

22	FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule	1988 – 2010s	G·D·K
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180	FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike	1992 – 2000s	G·K·N
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CLINICAL CARE

43	I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer	2014	H·K·N
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120	Bar-Code Medication Administration — Cue at the Point of Care	2010	H·K·D
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CLINICAL MEDICINE

95	Racial Bias in Pain Assessment — The False-Belief Mechanism	2016	T·K·N
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113	Removing the Race Coefficient from eGFR	2021	D·G·N
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CLINICAL/TRANSLATIONAL RESEARCH

178	Team Science Training for Clinical and Translational Scientists	2020 – 2022	K·N
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CORPORATE L&D

35	Training Transfer — The Gap Between Learning and Doing	2010	K·N
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46	High-Impact Learning System — Engineering the Environment, Not Just the Event	2001 – present	K·N
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108	The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops	1959 – present	K·N
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124	Brinkerhoff Success Case Method — Tails as the Evaluation Instrument	2005 – present	K·N
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CRIMINAL JUSTICE

99 COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate 2014 – 2018 **D-K-N**

DATA GOVERNANCE

190 CARE Principles — Indigenous Data Governance as a Replaced Regime 2019 – 2020 (principles); ongoing **G-N**

DIGITAL IDENTITY

187 Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India 2018 – 2025 **G-N-H**

E-GOVERNMENT

27 Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox) 2001 – present **G-K-N**

ED-TECH

96 Algorithmic Bias in Automated Exam Proctoring 2022 **D-N-K**

EDUCATION (NORWAY)

182 Norway's National Expert Commission on Learning Analytics 2022 – 2023 **G-K-N**

EDUCATION AT SCALE

40 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale 2016 **T-D**

100 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction 2012 – 2024 **D-K-N**

FINANCE

94 Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity 2018 – 2022 **D-G-N**

107 Fintech Lending Fairness Audit — When Including Race Reduces Disparity 2025 **D-G-N**

FINANCIAL SERVICES

163 Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model 2019 – 2021 **D-K-N**

GENERAL AVIATION

13 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010 **H-N-K**

GLOBAL HEALTH

47 PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption 2023 **K-N-D**

188 Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface 2013 – 2018 **H-N**

GOVERNMENT

98	UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days 2020	D·K·N
162	Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict 2016 – 2023	D·G·N
191	NYC Local Law 144 — Bias Audits as Governance Artifact 2023 – present	D·G·K

GOVERNMENT/WELFARE (NETHERLANDS)

155	Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds 2014 – 2020	D·G·N
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HEALTH PROFESSIONS EDUCATION

117	Interprofessional Education and the Evidence Gap 2013 – 2015	K·N
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HEALTHCARE/ONCOLOGY

67	Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018	D·K·N
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HIGHER EDUCATION

128	OU Analyse — Predictive Learning Analytics and Teacher Use at Scale 2019 – 2023	T·K·D
133	Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction 2024	D·K·N
157	Enrollment-Algorithm Yield Optimization Across U.S. Higher Education (Brookings synthesis 2021)	T·K·N
158	Crisis Point — Merit-Aid Leveraging at Public Flagships 2001 – 2017 (Burd IPEDS analysis); 2024 (Lifting the Veil)	T·K·N
159	GAO Online Program Manager Oversight Gap (GAO-22-104463) 2022	D·K·N
160	USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC) 2010s – 2024	D·K·N
161	In re 2U, Inc. Securities Class Action — When Yield Management Crashes Into Disclosure 2019 – 2022	D·K·N
170	Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions 2025	T·K·N

HIGHER EDUCATION (AFRICA)

169	Data Privacy and Learning Analytics on the African Continent 2022	K·G·N
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HIGHER EDUCATION (BRAZIL)

123	MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil) 2024	K·N
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HIGHER EDUCATION (LATIN AMERICA)

181	LALA — Building Learning-Analytics Governance Capacity Across Latin America	2017 – 2020	K·N
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HIGHER ED (UK)

115	Open University 'Ethical Use of Student Data' and OU Analyse	2014 – 2025	G·K·N
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HIGHER-ED ANALYTICS

106	Purdue Course Signals — The Reverse-Causality Retention Claim	2012 – 2013	D·K·N
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HOSPITAL PHARMACY

186	Australian Hospital-Pharmacy Technician Role Redesign	2016	D·N·H
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IMPLEMENTATION SCIENCE

179	Implementation-Science Training — Stated Goals Outrunning Operational Practice	2020s	K·N
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LEARNING ANALYTICS

50	The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension	2016 – 2025	T·K·D
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134	Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models	2021 – 2024	D·K·N
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193	Learning Analytics on the African Continent — A Scoping Review as the Present-State Map	2022	K·N
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MEDICAL DEVICES

114	Pulse Oximetry Across Skin Tones	1990 – 2020 – 2025	D·G·N
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MEDICAL EDUCATION

41	Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA	2009 – 2020s	T·K
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45	Spaced Education RCTs in Medical Training	2007 – 2009	H·K·D
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MEDICAL-DEVICE REGULATION

189	Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD	2014 – present	G·N
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MULTIMODAL LEARNING ANALYTICS

53	Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account	2023	T·K·N
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NAVAL ENGINEERING

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|-----------|---|----------------|--------------|
| 19 | Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable | 1963 – present | G·K·H |
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NEUROSCIENCE

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| 16 | EU Human Brain Project — Top-Down Vision That Unraveled | 2013 – 2023 | G·N |
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| 176 | Launching the BRAIN Initiative — Governance of a Grand Challenge | 2011 – 2015 – present | G·K·N |
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NEUROSCIENCE/CONNECTOMICS

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| 81 | CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale | 2017 – present | H·K·N |
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NUCLEAR ENGINEERING

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| 76 | INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet | 2010 – present | D·H·K |
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NUCLEAR POWER

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| 26 | INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding | 2014 | G·K·H |
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NURSING EDUCATION

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| 44 | NCSBN National Simulation Study — Licensing the 50% Substitution Rule | 2014 | G·K·D |
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PHARMA SAFETY

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| 168 | iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms | 2006 – 2011 | G·D·N |
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RAIL TRANSPORT

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| 177 | CIRAS — Confidential Incident Reporting for UK Rail | 1996 – present | G·K·N |
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SOFTWARE ENGINEERING EDUCATION

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| 48 | Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development | 2025 (preprint) | K·N |
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SOFTWARE SUSTAINMENT

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| 24 | Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed | 1996 – 2000 | G·D·K |
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SPACE / AEROSPACE

STANDOUT FAILURE	Case 140	Challenger & Columbia (1986 / 2003). Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.
NOTE	Case 18	No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 18) for the safety-engineering counterpart that aerospace did not build.

9	Mars Climate Orbiter — Unit Mismatch	1999	D-K
15	Ariane 5 Flight 501	1996	D-K-H
65	NASA EVA-23 — Water Intrusion Inside a Spacesuit	2013	H-N-D
140	Challenger & Columbia	1986 / 2003	N-K-G
165	Boeing Starliner	2019 – 2024	K-D

SURGERY

121	Surgical Skill Video Peer-Rating Predicts Complications	2013	H-D-K
122	Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units	2009 – 2016	T-K-H

TECHNOLOGY

29	Amazon Hiring AI — Trained Bias, Deprecated 2018	2014 – 2018	D-K-N
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TUTORING

77	Hybrid Human-AI Tutoring — Augmentation, Not Delegation	2024	T-K-D
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WORKFORCE DEV

39	CIRCUIT — A Scalable, Equity-Centered Research Workforce Model	2017 – 2023 (six cycles)	T-K
185	Singapore SkillsFuture — National Workforce Capability at Scale	2015 – present	G-K-D

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

34 cases

1 USS Fitzgerald & USS John S. McCain

7 Boeing 737 MAX / MCAS

14 Healthcare.gov Launch

16 EU Human Brain Project — Top-Down Vision That Unraveled

17 GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data

19 Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable

21 MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

22 FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule

23 FAA NextGen and the ADS-B Out Transition

24 Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed

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 - 62 Uber ATG / Tempe Fatality

 - 63 Eastern Air Lines Flight 401

 - 64 Stanislav Petrov / 1983 False Alert

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51 cases

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Learning Engineering for Next-Generation Systems · v2.1 · Edition · 15 June 2026

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